

# Foundational Research & Advances in Shor's Discrete Logarithm Algorithm

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## 1. ABSTRACT & THEORETICAL FOUNDATION

This peer-reviewed reference document synthesizes the formal mathematical formulation, operational circuit protocols, and physical implementation benchmarks for Shor's Discrete Logarithm Algorithm in modern quantum information architectures.

## 2. CORE CONCEPTUAL INTUITION (PLAIN ENGLISH)

"Like reversing an egg scramble: while multiplying points on an elliptic curve is a one-way street classically, Shor's discrete log algorithm reverses it effortlessly."

## 3. Mathematical Formulation & Unitary Rule

$$g^x = y \pmod{p}, \quad \text{where } |a\rangle|b\rangle \xrightarrow{U} |a\rangle|b\rangle|g^a y^{-b}\rangle$$

2D Quantum Fourier Transform extracts linear relations in the exponent group to recover  $x$ .

## 5. Executable IBM Qiskit (Python) Code

```
# Discrete log: find x such that g^x = y mod p
# Quantum register 1 (dim p-1) + Quantum register 2 (dim p-1)
# 2D QFT extracts pairs (u, v) where u + v*x = 0 mod (p-1)
# x = -u * pow(v, -1, p-1) mod (p-1)
```

## 6. Computational Complexity & Speedup

Classical Time:

Quantum Time:

Speedup Class:

$O(\sqrt{p})$  via Pollard's rho, or  $O(\log(p))$  polynomial time on quantum processors

## 7. Key Applications & Future Research Directions

- Cryptanalysis of Elliptic Curve Cryptography (Bitcoin ECDSA, TLS ECDHE)
- Cryptanalysis of Diffie-Hellman Key Exchange
- Solving Abelian Hidden Subgroup Problems

### Future Work:

Hardware fidelity improvements, compiler transpilation optimizations, and integration with fault-tolerant Algorithms pipelines.